Results for Eedi

New held-out split; N = 5, 10, ..., 50.

Metrics: Adaptability (Ada), Novelty (NOV), and Ep at thresholds 0.4, 0.5, 0.6, 0.7.

Cell format: mean +/- sample SD; deterministic CF baselines show one value.

Bold = best; underline = second best by unrounded mean.

Ablation methods
2CKG4ER
No-mastery
No-forgetting
No-student-feature
No-exercise-feature
ID-based only

Comparison methods
EB-CF
SB-CF
CBF
TransE
TransE-adv
RotatE
DistMult
ComplEx
2CKG4ER

Experimental setting

All methods use the selected drop-zero recommendation graph for this dataset.

Graph parameters: theta_m = 0.8, theta_f = 0.8, rec = 50.

Source: Data_Fin_New experiment metrics; all methods included.

Ablation performance

Six variants evaluated on the selected graph.

(a) Adaptability (Ada)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	0.700 ± 0.002	0.681 ± 0.003	**0.713 ± 0.001**	0.688 ± 0.005	0.699 ± 0.002	0.689 ± 0.001
10	0.700 ± 0.002	0.682 ± 0.004	**0.706 ± 0.000**	0.688 ± 0.004	0.699 ± 0.001	0.689 ± 0.002
15	0.699 ± 0.002	0.681 ± 0.003	**0.701 ± 0.005**	0.687 ± 0.003	0.698 ± 0.001	0.690 ± 0.002
20	0.698 ± 0.001	0.681 ± 0.002	**0.700 ± 0.003**	0.687 ± 0.003	0.698 ± 0.000	0.689 ± 0.002
25	0.698 ± 0.001	0.682 ± 0.002	**0.698 ± 0.001**	0.687 ± 0.003	0.697 ± 0.000	0.689 ± 0.002
30	0.697 ± 0.001	0.682 ± 0.001	**0.697 ± 0.000**	0.687 ± 0.003	0.697 ± 0.000	0.689 ± 0.002
35	**0.697 ± 0.000**	0.682 ± 0.001	0.696 ± 0.000	0.686 ± 0.003	0.696 ± 0.000	0.689 ± 0.001
40	**0.696 ± 0.000**	0.682 ± 0.001	0.696 ± 0.000	0.686 ± 0.002	0.696 ± 0.000	0.689 ± 0.001
45	**0.696 ± 0.000**	0.683 ± 0.000	0.696 ± 0.000	0.686 ± 0.002	0.696 ± 0.000	0.688 ± 0.001
50	**0.696 ± 0.000**	0.683 ± 0.001	0.695 ± 0.000	0.686 ± 0.002	0.695 ± 0.000	0.688 ± 0.001

(b) Novelty (NOV)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	**0.990 ± 0.002**	0.985 ± 0.003	0.984 ± 0.001	0.986 ± 0.001	0.986 ± 0.002	0.982 ± 0.003
10	**0.989 ± 0.002**	0.986 ± 0.003	0.981 ± 0.000	0.985 ± 0.000	0.987 ± 0.001	0.982 ± 0.003
15	**0.987 ± 0.002**	0.985 ± 0.001	0.984 ± 0.001	0.985 ± 0.001	0.987 ± 0.001	0.981 ± 0.002
20	**0.987 ± 0.001**	0.985 ± 0.001	0.985 ± 0.000	0.984 ± 0.001	0.986 ± 0.001	0.980 ± 0.002
25	0.986 ± 0.001	0.985 ± 0.001	**0.986 ± 0.000**	0.983 ± 0.001	0.984 ± 0.002	0.981 ± 0.001
30	**0.986 ± 0.000**	0.985 ± 0.001	0.986 ± 0.000	0.982 ± 0.001	0.984 ± 0.002	0.981 ± 0.001
35	0.985 ± 0.001	0.984 ± 0.001	**0.986 ± 0.001**	0.982 ± 0.001	0.984 ± 0.002	0.981 ± 0.000
40	0.985 ± 0.000	0.984 ± 0.001	**0.986 ± 0.001**	0.982 ± 0.002	0.984 ± 0.001	0.981 ± 0.001
45	0.984 ± 0.001	0.984 ± 0.001	**0.985 ± 0.001**	0.981 ± 0.002	0.983 ± 0.001	0.982 ± 0.000
50	0.984 ± 0.001	0.984 ± 0.001	**0.985 ± 0.001**	0.981 ± 0.003	0.983 ± 0.001	0.982 ± 0.000

(theta_m, theta_f, rec) = (0.8, 0.8, 50); zero exercise-forgetting edges removed.

Ablation performance

Six variants evaluated on the selected graph.

(c) Ep, threshold 0.4

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	0.014 ± 0.105	**0.206 ± 0.092**	0.089 ± 0.064	0.104 ± 0.007	0.055 ± 0.105	0.117 ± 0.013
10	0.043 ± 0.089	**0.312 ± 0.061**	0.081 ± 0.024	0.182 ± 0.010	0.079 ± 0.091	0.191 ± 0.020
15	0.103 ± 0.155	**0.374 ± 0.077**	0.214 ± 0.132	0.228 ± 0.001	0.124 ± 0.098	0.245 ± 0.016
20	0.145 ± 0.081	**0.375 ± 0.087**	0.305 ± 0.124	0.262 ± 0.006	0.147 ± 0.032	0.287 ± 0.024
25	0.190 ± 0.095	0.390 ± 0.116	**0.433 ± 0.038**	0.286 ± 0.010	0.192 ± 0.052	0.310 ± 0.027
30	0.252 ± 0.068	0.419 ± 0.068	**0.512 ± 0.016**	0.307 ± 0.009	0.221 ± 0.016	0.333 ± 0.019
35	0.337 ± 0.137	0.452 ± 0.064	**0.524 ± 0.020**	0.324 ± 0.010	0.252 ± 0.079	0.347 ± 0.027
40	0.352 ± 0.072	0.434 ± 0.068	**0.533 ± 0.044**	0.345 ± 0.014	0.297 ± 0.115	0.359 ± 0.024
45	0.412 ± 0.009	0.398 ± 0.103	**0.534 ± 0.029**	0.360 ± 0.010	0.353 ± 0.125	0.378 ± 0.024
50	0.495 ± 0.049	0.395 ± 0.117	**0.515 ± 0.006**	0.370 ± 0.019	0.380 ± 0.093	0.395 ± 0.026

(d) Ep, threshold 0.5

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	-0.054 ± 0.097	**0.130 ± 0.086**	0.018 ± 0.067	0.021 ± 0.008	-0.028 ± 0.096	0.027 ± 0.013
10	-0.102 ± 0.093	**0.175 ± 0.062**	-0.097 ± 0.013	0.030 ± 0.009	-0.065 ± 0.103	0.032 ± 0.027
15	-0.124 ± 0.144	**0.172 ± 0.089**	-0.042 ± 0.113	0.019 ± 0.012	-0.083 ± 0.108	0.033 ± 0.027
20	-0.151 ± 0.098	**0.130 ± 0.074**	-0.010 ± 0.122	0.005 ± 0.020	-0.131 ± 0.066	0.032 ± 0.022
25	-0.151 ± 0.122	0.084 ± 0.080	**0.084 ± 0.068**	-0.011 ± 0.030	-0.133 ± 0.100	0.016 ± 0.024
30	-0.120 ± 0.076	0.069 ± 0.099	**0.177 ± 0.063**	-0.020 ± 0.032	-0.146 ± 0.071	0.005 ± 0.027
35	-0.077 ± 0.118	0.089 ± 0.108	**0.187 ± 0.031**	-0.028 ± 0.032	-0.152 ± 0.154	-0.008 ± 0.035
40	-0.118 ± 0.054	0.059 ± 0.123	**0.218 ± 0.040**	-0.032 ± 0.036	-0.153 ± 0.177	-0.011 ± 0.032
45	-0.091 ± 0.038	0.017 ± 0.145	**0.248 ± 0.045**	-0.040 ± 0.036	-0.150 ± 0.174	-0.012 ± 0.037
50	-0.033 ± 0.030	0.014 ± 0.166	**0.264 ± 0.034**	-0.047 ± 0.040	-0.155 ± 0.163	-0.008 ± 0.040

(theta_m, theta_f, rec) = (0.8, 0.8, 50); zero exercise-forgetting edges removed.

Ablation performance

Six variants evaluated on the selected graph.

(e) Ep, threshold 0.6

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	-0.116 ± 0.081	**0.028 ± 0.080**	-0.065 ± 0.091	-0.052 ± 0.002	-0.090 ± 0.097	-0.052 ± 0.009
10	-0.246 ± 0.116	**0.005 ± 0.067**	-0.295 ± 0.038	-0.101 ± 0.006	-0.198 ± 0.114	-0.114 ± 0.034
15	-0.350 ± 0.179	**-0.080 ± 0.077**	-0.291 ± 0.121	-0.169 ± 0.010	-0.304 ± 0.138	-0.164 ± 0.046
20	-0.462 ± 0.148	**-0.181 ± 0.050**	-0.311 ± 0.136	-0.223 ± 0.014	-0.421 ± 0.116	-0.211 ± 0.040
25	-0.549 ± 0.145	-0.272 ± 0.051	-0.273 ± 0.204	-0.274 ± 0.028	-0.489 ± 0.096	**-0.264 ± 0.045**
30	-0.593 ± 0.089	-0.329 ± 0.077	**-0.217 ± 0.213**	-0.313 ± 0.032	-0.554 ± 0.045	-0.302 ± 0.053
35	-0.576 ± 0.120	-0.403 ± 0.117	**-0.238 ± 0.176**	-0.350 ± 0.040	-0.630 ± 0.115	-0.335 ± 0.063
40	-0.660 ± 0.066	-0.487 ± 0.154	**-0.258 ± 0.144**	-0.372 ± 0.052	-0.688 ± 0.115	-0.362 ± 0.065
45	-0.692 ± 0.025	-0.574 ± 0.176	**-0.302 ± 0.180**	-0.399 ± 0.056	-0.717 ± 0.111	-0.381 ± 0.074
50	-0.654 ± 0.017	-0.641 ± 0.171	**-0.338 ± 0.162**	-0.424 ± 0.066	-0.753 ± 0.093	-0.397 ± 0.083

(f) Ep, threshold 0.7

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	-0.180 ± 0.066	**-0.060 ± 0.076**	-0.141 ± 0.092	-0.132 ± 0.005	-0.174 ± 0.100	-0.126 ± 0.007
10	-0.376 ± 0.097	**-0.162 ± 0.088**	-0.452 ± 0.042	-0.259 ± 0.004	-0.349 ± 0.090	-0.262 ± 0.018
15	-0.517 ± 0.134	**-0.345 ± 0.043**	-0.508 ± 0.097	-0.382 ± 0.007	-0.518 ± 0.091	-0.381 ± 0.021
20	-0.656 ± 0.097	-0.543 ± 0.035	-0.582 ± 0.111	-0.482 ± 0.004	-0.652 ± 0.026	**-0.479 ± 0.030**
25	-0.739 ± 0.094	-0.652 ± 0.027	**-0.563 ± 0.056**	-0.572 ± 0.016	-0.731 ± 0.010	-0.564 ± 0.034
30	-0.803 ± 0.093	-0.712 ± 0.101	**-0.535 ± 0.040**	-0.642 ± 0.016	-0.792 ± 0.018	-0.625 ± 0.031
35	-0.792 ± 0.140	-0.772 ± 0.060	**-0.600 ± 0.017**	-0.695 ± 0.011	-0.815 ± 0.059	-0.675 ± 0.038
40	-0.852 ± 0.078	-0.862 ± 0.066	**-0.672 ± 0.094**	-0.737 ± 0.022	-0.828 ± 0.064	-0.707 ± 0.043
45	-0.844 ± 0.025	-0.899 ± 0.067	**-0.724 ± 0.081**	-0.770 ± 0.025	-0.841 ± 0.040	-0.729 ± 0.047
50	-0.794 ± 0.045	-0.905 ± 0.089	-0.776 ± 0.056	-0.796 ± 0.036	-0.850 ± 0.028	**-0.748 ± 0.048**

(theta_m, theta_f, rec) = (0.8, 0.8, 50); zero exercise-forgetting edges removed.

Comparison performance

Eight baselines and 2CKG4ER evaluated on the selected graph.

(a) Adaptability (Ada)

N	EB-CF	SB-CF	CBF	TransE	TransE-adv	RotatE	DistMult	ComplEx	2CKG4ER
5	0.688	0.689	0.612	0.689 ± 0.002	0.691 ± 0.002	0.685 ± 0.001	0.687 ± 0.003	0.689 ± 0.002	**0.700 ± 0.002**
10	0.688	0.688	0.616	0.690 ± 0.002	0.692 ± 0.002	0.686 ± 0.000	0.688 ± 0.003	0.690 ± 0.002	**0.700 ± 0.002**
15	0.688	0.688	0.619	0.690 ± 0.001	0.692 ± 0.002	0.686 ± 0.000	0.688 ± 0.002	0.690 ± 0.001	**0.699 ± 0.002**
20	0.688	0.687	0.621	0.690 ± 0.001	0.692 ± 0.002	0.686 ± 0.001	0.688 ± 0.002	0.690 ± 0.001	**0.698 ± 0.001**
25	0.688	0.687	0.624	0.690 ± 0.001	0.692 ± 0.001	0.687 ± 0.001	0.689 ± 0.002	0.690 ± 0.001	**0.698 ± 0.001**
30	0.688	0.687	0.626	0.689 ± 0.001	0.692 ± 0.001	0.687 ± 0.001	0.689 ± 0.002	0.690 ± 0.001	**0.697 ± 0.001**
35	0.688	0.687	0.628	0.689 ± 0.001	0.691 ± 0.001	0.687 ± 0.001	0.689 ± 0.002	0.690 ± 0.001	**0.697 ± 0.000**
40	0.688	0.687	0.630	0.689 ± 0.001	0.691 ± 0.001	0.687 ± 0.001	0.689 ± 0.001	0.690 ± 0.001	**0.696 ± 0.000**
45	0.688	0.687	0.632	0.689 ± 0.001	0.691 ± 0.001	0.688 ± 0.001	0.689 ± 0.001	0.690 ± 0.000	**0.696 ± 0.000**
50	0.688	0.687	0.633	0.689 ± 0.001	0.691 ± 0.001	0.688 ± 0.001	0.689 ± 0.001	0.690 ± 0.000	**0.696 ± 0.000**

(b) Novelty (NOV)

N	EB-CF	SB-CF	CBF	TransE	TransE-adv	RotatE	DistMult	ComplEx	2CKG4ER
5	0.986	0.980	0.999	0.999 ± 0.000	0.999 ± 0.001	**0.999 ± 0.000**	0.997 ± 0.001	0.999 ± 0.001	0.990 ± 0.002
10	0.988	0.983	**0.999**	0.999 ± 0.000	0.998 ± 0.001	0.999 ± 0.000	0.997 ± 0.001	0.999 ± 0.001	0.989 ± 0.002
15	0.986	0.982	**0.999**	0.999 ± 0.001	0.998 ± 0.001	0.999 ± 0.000	0.997 ± 0.001	0.998 ± 0.001	0.987 ± 0.002
20	0.985	0.981	**0.999**	0.998 ± 0.001	0.997 ± 0.002	0.999 ± 0.001	0.997 ± 0.001	0.998 ± 0.001	0.987 ± 0.001
25	0.985	0.982	**0.999**	0.997 ± 0.001	0.997 ± 0.002	0.999 ± 0.000	0.997 ± 0.001	0.998 ± 0.001	0.986 ± 0.001
30	0.985	0.981	**0.999**	0.996 ± 0.001	0.997 ± 0.002	0.999 ± 0.000	0.997 ± 0.001	0.998 ± 0.001	0.986 ± 0.000
35	0.984	0.981	**0.999**	0.996 ± 0.001	0.996 ± 0.001	0.999 ± 0.000	0.997 ± 0.000	0.998 ± 0.001	0.985 ± 0.001
40	0.984	0.981	**0.999**	0.995 ± 0.001	0.996 ± 0.001	0.999 ± 0.000	0.997 ± 0.000	0.998 ± 0.001	0.985 ± 0.000
45	0.984	0.981	**0.999**	0.995 ± 0.001	0.996 ± 0.001	0.999 ± 0.000	0.997 ± 0.000	0.998 ± 0.001	0.984 ± 0.001
50	0.984	0.981	**0.999**	0.995 ± 0.001	0.995 ± 0.001	0.999 ± 0.000	0.997 ± 0.000	0.998 ± 0.001	0.984 ± 0.001

(theta_m, theta_f, rec) = (0.8, 0.8, 50); zero exercise-forgetting edges removed.

Comparison performance

Eight baselines and 2CKG4ER evaluated on the selected graph.

(c) Ep, threshold 0.4

N	EB-CF	SB-CF	CBF	TransE	TransE-adv	RotatE	DistMult	ComplEx	2CKG4ER
5	0.029	0.099	0.114	0.118 ± 0.008	0.110 ± 0.008	**0.118 ± 0.007**	0.118 ± 0.004	0.118 ± 0.008	0.014 ± 0.105
10	0.066	0.176	**0.205**	0.187 ± 0.019	0.184 ± 0.017	0.199 ± 0.015	0.195 ± 0.006	0.191 ± 0.013	0.043 ± 0.089
15	0.114	0.234	0.245	0.228 ± 0.021	0.225 ± 0.016	**0.263 ± 0.008**	0.235 ± 0.006	0.233 ± 0.004	0.103 ± 0.155
20	0.144	0.296	0.280	0.243 ± 0.014	0.252 ± 0.018	**0.297 ± 0.005**	0.271 ± 0.004	0.265 ± 0.010	0.145 ± 0.081
25	0.184	**0.335**	0.296	0.255 ± 0.019	0.275 ± 0.021	0.325 ± 0.002	0.300 ± 0.012	0.294 ± 0.017	0.190 ± 0.095
30	0.225	0.348	0.311	0.262 ± 0.026	0.300 ± 0.029	**0.353 ± 0.007**	0.326 ± 0.012	0.318 ± 0.027	0.252 ± 0.068
35	0.242	**0.371**	0.333	0.266 ± 0.028	0.313 ± 0.025	0.364 ± 0.010	0.344 ± 0.014	0.327 ± 0.021	0.337 ± 0.137
40	0.269	**0.380**	0.365	0.283 ± 0.031	0.319 ± 0.030	0.379 ± 0.003	0.351 ± 0.014	0.332 ± 0.013	0.352 ± 0.072
45	0.282	0.385	0.389	0.297 ± 0.036	0.325 ± 0.030	0.383 ± 0.004	0.361 ± 0.005	0.347 ± 0.014	**0.412 ± 0.009**
50	0.298	0.402	0.403	0.310 ± 0.039	0.340 ± 0.016	0.384 ± 0.015	0.377 ± 0.006	0.355 ± 0.019	**0.495 ± 0.049**

(d) Ep, threshold 0.5

N	EB-CF	SB-CF	CBF	TransE	TransE-adv	RotatE	DistMult	ComplEx	2CKG4ER
5	-0.002	0.024	0.030	**0.039 ± 0.004**	0.024 ± 0.017	0.038 ± 0.002	0.025 ± 0.007	0.025 ± 0.006	-0.054 ± 0.097
10	0.004	0.023	0.045	0.036 ± 0.022	0.024 ± 0.027	**0.048 ± 0.012**	0.043 ± 0.010	0.032 ± 0.011	-0.102 ± 0.093
15	0.030	0.039	0.043	0.022 ± 0.026	0.011 ± 0.025	**0.056 ± 0.006**	0.039 ± 0.017	0.028 ± 0.011	-0.124 ± 0.144
20	0.048	**0.064**	0.056	-0.000 ± 0.024	-0.002 ± 0.027	0.041 ± 0.007	0.030 ± 0.014	0.016 ± 0.016	-0.151 ± 0.098
25	0.072	**0.077**	0.039	-0.015 ± 0.025	-0.010 ± 0.027	0.033 ± 0.003	0.025 ± 0.018	0.010 ± 0.015	-0.151 ± 0.122
30	**0.107**	0.075	0.026	-0.029 ± 0.029	-0.011 ± 0.042	0.038 ± 0.005	0.019 ± 0.014	0.006 ± 0.017	-0.120 ± 0.076
35	**0.111**	0.092	0.004	-0.040 ± 0.029	-0.016 ± 0.039	0.027 ± 0.017	0.019 ± 0.007	-0.004 ± 0.015	-0.077 ± 0.118
40	**0.128**	0.098	0.017	-0.039 ± 0.032	-0.029 ± 0.047	0.023 ± 0.024	0.011 ± 0.007	-0.022 ± 0.013	-0.118 ± 0.054
45	**0.128**	0.095	0.012	-0.044 ± 0.035	-0.044 ± 0.044	0.012 ± 0.029	0.009 ± 0.000	-0.030 ± 0.019	-0.091 ± 0.038
50	**0.135**	0.095	0.011	-0.047 ± 0.038	-0.047 ± 0.033	0.001 ± 0.038	0.017 ± 0.008	-0.035 ± 0.024	-0.033 ± 0.030

(theta_m, theta_f, rec) = (0.8, 0.8, 50); zero exercise-forgetting edges removed.

Comparison performance

Eight baselines and 2CKG4ER evaluated on the selected graph.

(e) Ep, threshold 0.6

N	EB-CF	SB-CF	CBF	TransE	TransE-adv	RotatE	DistMult	ComplEx	2CKG4ER
5	-0.048	-0.062	-0.042	**-0.033 ± 0.017**	-0.051 ± 0.014	-0.037 ± 0.007	-0.051 ± 0.001	-0.051 ± 0.002	-0.116 ± 0.081
10	**-0.076**	-0.122	-0.094	-0.085 ± 0.026	-0.106 ± 0.031	-0.094 ± 0.005	-0.091 ± 0.014	-0.102 ± 0.009	-0.246 ± 0.116
15	**-0.085**	-0.159	-0.156	-0.133 ± 0.034	-0.151 ± 0.031	-0.141 ± 0.006	-0.136 ± 0.025	-0.154 ± 0.014	-0.350 ± 0.179
20	**-0.102**	-0.176	-0.210	-0.179 ± 0.033	-0.190 ± 0.037	-0.203 ± 0.011	-0.181 ± 0.014	-0.207 ± 0.020	-0.462 ± 0.148
25	**-0.109**	-0.187	-0.253	-0.209 ± 0.037	-0.229 ± 0.046	-0.241 ± 0.007	-0.226 ± 0.015	-0.252 ± 0.019	-0.549 ± 0.145
30	**-0.104**	-0.209	-0.306	-0.243 ± 0.041	-0.256 ± 0.068	-0.271 ± 0.005	-0.263 ± 0.014	-0.282 ± 0.015	-0.593 ± 0.089
35	**-0.133**	-0.215	-0.343	-0.272 ± 0.032	-0.288 ± 0.068	-0.308 ± 0.007	-0.292 ± 0.012	-0.324 ± 0.019	-0.576 ± 0.120
40	**-0.157**	-0.226	-0.357	-0.292 ± 0.023	-0.321 ± 0.073	-0.332 ± 0.008	-0.321 ± 0.010	-0.366 ± 0.015	-0.660 ± 0.066
45	**-0.187**	-0.242	-0.378	-0.311 ± 0.029	-0.343 ± 0.067	-0.361 ± 0.003	-0.342 ± 0.014	-0.390 ± 0.014	-0.692 ± 0.025
50	**-0.213**	-0.266	-0.401	-0.327 ± 0.034	-0.358 ± 0.050	-0.384 ± 0.018	-0.358 ± 0.007	-0.413 ± 0.015	-0.654 ± 0.017

(f) Ep, threshold 0.7

N	EB-CF	SB-CF	CBF	TransE	TransE-adv	RotatE	DistMult	ComplEx	2CKG4ER
5	-0.117	-0.146	**-0.108**	-0.117 ± 0.019	-0.119 ± 0.007	-0.113 ± 0.008	-0.120 ± 0.009	-0.125 ± 0.002	-0.180 ± 0.066
10	**-0.203**	-0.269	-0.217	-0.224 ± 0.025	-0.219 ± 0.025	-0.237 ± 0.014	-0.218 ± 0.018	-0.242 ± 0.007	-0.376 ± 0.097
15	**-0.257**	-0.364	-0.332	-0.313 ± 0.025	-0.306 ± 0.033	-0.337 ± 0.004	-0.306 ± 0.008	-0.339 ± 0.011	-0.517 ± 0.134
20	**-0.299**	-0.398	-0.432	-0.389 ± 0.012	-0.382 ± 0.050	-0.428 ± 0.006	-0.388 ± 0.012	-0.429 ± 0.016	-0.656 ± 0.097
25	**-0.333**	-0.433	-0.516	-0.445 ± 0.020	-0.440 ± 0.054	-0.492 ± 0.006	-0.458 ± 0.003	-0.499 ± 0.020	-0.739 ± 0.094
30	**-0.360**	-0.480	-0.595	-0.492 ± 0.024	-0.484 ± 0.069	-0.531 ± 0.003	-0.503 ± 0.007	-0.552 ± 0.031	-0.803 ± 0.093
35	**-0.437**	-0.509	-0.640	-0.523 ± 0.018	-0.525 ± 0.061	-0.574 ± 0.004	-0.548 ± 0.010	-0.606 ± 0.037	-0.792 ± 0.140
40	**-0.499**	-0.546	-0.659	-0.550 ± 0.021	-0.560 ± 0.061	-0.604 ± 0.006	-0.592 ± 0.014	-0.652 ± 0.034	-0.852 ± 0.078
45	**-0.560**	-0.582	-0.683	-0.577 ± 0.020	-0.589 ± 0.052	-0.637 ± 0.006	-0.626 ± 0.023	-0.689 ± 0.038	-0.844 ± 0.025
50	-0.613	-0.627	-0.706	**-0.602 ± 0.022**	-0.603 ± 0.038	-0.668 ± 0.011	-0.649 ± 0.017	-0.720 ± 0.038	-0.794 ± 0.045

(theta_m, theta_f, rec) = (0.8, 0.8, 50); zero exercise-forgetting edges removed.